

Thevenin's Theorem Lab Procedure

Instructional Objectives

1. To work through the procedural steps involved in Thevenin's theorem.
2. To verify the values obtained by measuring them using the digital multimeter.
3. To construct a Thevenin equivalent circuit.

Procedure

0. Parts: 1-3.3K Ω , 1-6.2K Ω and 1-8.2K Ω .

We will run through 12 steps to use Thevenin's theorem to find the current through R3 shown in Fig. 5.

1. Building the circuit and doing measurements:

a. Measure R1, R2 and R3 using Digital Multi Meter (DMM)

R1 = 3.3128 K Ω , R2 = 8.077 K Ω , R3 = 6.131 K Ω

b. Setup the (YELLOW) power supply to deliver 5V DC with a 25mA current limit to the breadboard.

c. Build the circuit as shown in Figure 5.

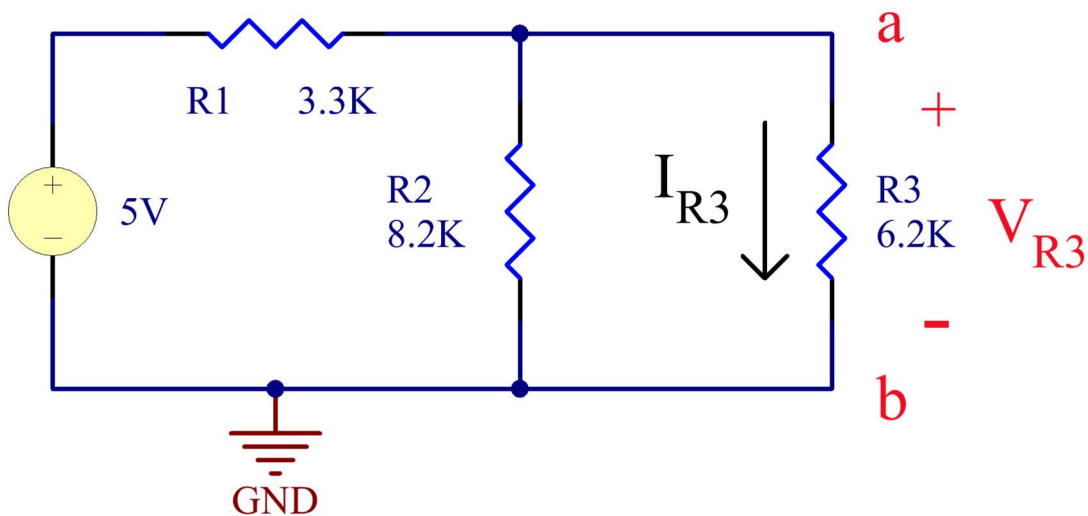


Figure 5

d. Measure the voltage across R3 (V_{R3}) using the DMM. Record the voltage and use the measured value of R3 to **Calculate** I_{R3} :

V_{R3} = 2.5633 V

$I_{R3} = V_{R3} / R3 =$ 0.000418088 A

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2. Analyzing the circuit.

- a. Referring to Fig. 6, which is Fig. 5 with R3 removed, calculate V_{TH} for Fig.6, **showing/describing all work.**

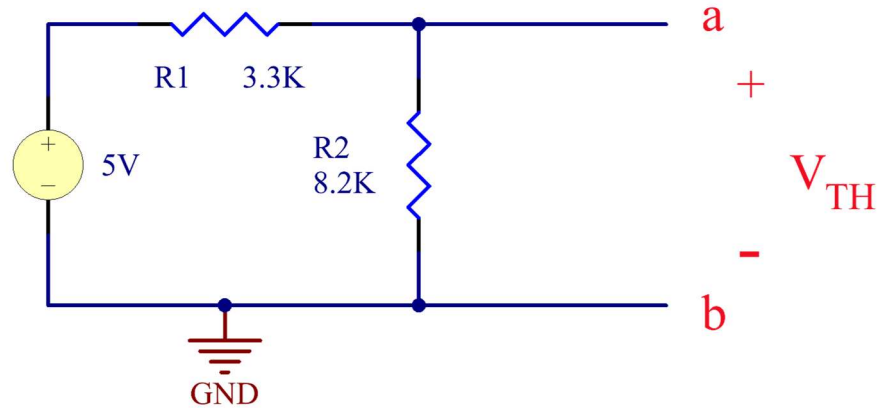


Figure 6

Note: $V_{th} = V_a$.

Simple Voltage divider $\rightarrow V_{th} = ((R_2)/(R_{tot})) * V_s$

$$R_{tot} = 8200 + 3300 = 11500$$

$$V_{th} = (8200/11500) * 5 = 3.5652 \text{ V}$$

b. $V_{TH} =$ 3.5652 V (calculated)

- c. Verify the real Thevenin equivalent voltage by measurement. Construct the circuit shown in Fig.6. Measure and record V_{TH} .

$V_{TH} =$ 3.5452 V (measured)

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3. Measuring R_{TH} by removing the source and replacing it with a short circuit.

- a. Construct the circuit in Fig. 7, which is the circuit in Fig. 5 with R_3 removed and the 5V source replaced by a short circuit (R_{SOURCE} assumed 0).

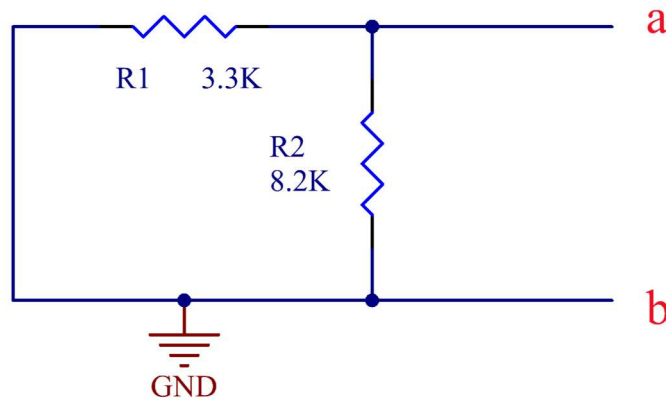


Figure 7

- b. Calculate R_{TH} in Fig. 7, **showing/demonstrating all work.**

$R_{th} = 3300 \parallel 8200$, this is because the path a->b would see these two resistors as parallel as they share the same two nodes.

$$R_{th} = ((1/3300 + 1/8200) ^{-1})$$

$$R_{th} = 2353.0435 \text{ ohms}$$

$$R_{TH} = \underline{2535.0435 \text{ ohms}} \text{ (calculated)}$$

- c. Verify your R_{TH} calculation by measurement. Connect the circuit in Fig. 7. Measure and record the equivalent resistance (R_{TH}) measured between terminals A and B.

$$R_{TH} = \underline{2349.8 \text{ ohms}} \text{ (measured)}$$

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4. Drawing the Thevenin equivalent circuit.

- a. Draw the Thevenin equivalent circuit and use your calculated values for V_{TH} and R_{TH} . **Label this Fig. 8.** You can use LTSpice or any other program to draw the circuit. Even a photo of a hand drawn picture.

Figure 8

- b. With the Thevenin equivalent circuit loaded with R_3 , calculate I_{R3} using the Thevenin equivalent circuit from Fig. 8.

$$I_{R3} = \frac{V_{TH}}{R_{TH} + R_3} \quad (6) \quad I_{R3} = (3.565)/(2353.04 + 6200) = 0.4168 \text{ mA}$$

- c. Compare the current calculated in procedure 1.d and the current calculated in procedure 4.b. Find the reasons for any discrepancy.

1.d: 0.41808 mA VS 4.b: 0.4168 mA

The two values of I_{R3} differ by less than 0.002 mA, an order of magnitude that certainly pertains to noise and precision of numbers used in calculation than anything else. Reasons for discrepancy could be analog noise, imperfect internal resistance/resistor values, and calculation precision.

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5. Building Thevenin equivalent circuit

a. Build the circuit of Fig. 8. Obtain a resistance for R_{TH} as close as possible to its calculated value. It doesn't have to be 1 resistor. Hint 2 resistors in parallel is 'real close'.

b. Measure the value of R_{TH} and V_{TH} using the DMM.

$$R_{TH} = \underline{2350} \, \Omega \quad V_{TH} = \underline{3.5658} \, V$$

***** Load the Thevenin equivalent circuit with R3. *****

c. Measure the voltage across R_{TH} , (V_{RTH} not V_{TH}).

$$V_{RTH} = \underline{0.98756} \, V$$

d. Measure the voltage across the load resistor R_3 (V_3)

$$V_{R3} = \underline{2.5773} \, V$$

e. Calculate the current through R_{TH} and R_3 using measured values of V_{TH} , R_{TH} and R_3 (The resistors are in series so the same current flows through each of them).

$$I = I_{R3} = V_{RTH} / R_{TH} = V_{TH} / (R_{TH} + R_3) \underline{0.42044} \, mA$$

f. Compare these measured results with the results of steps 4. Find the reasons for any discrepancies.

g. Compare the current calculated in procedure 1.d and the current calculated in procedure 5.e. Find the reasons for any discrepancy

5.f&g: This value is more off when compared with the other two, but still on an order of magnitude that is negligible when questioning theoretical convergence. The different values I have measured for R_{th} and V_{th} have differed slightly so this may be the reason. Again, obvious reasons for discrepancy can be analog noise, imperfect internal resistance/resistor values, and calculation precision. In any case, the values from 1.d, 4b, and 5.e are all in accordance with each other and what is reasonable error.

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6. Complex Thevenin equivalent circuit

In this step, you are required to apply the Thevenin theorem to a more complicated circuit.

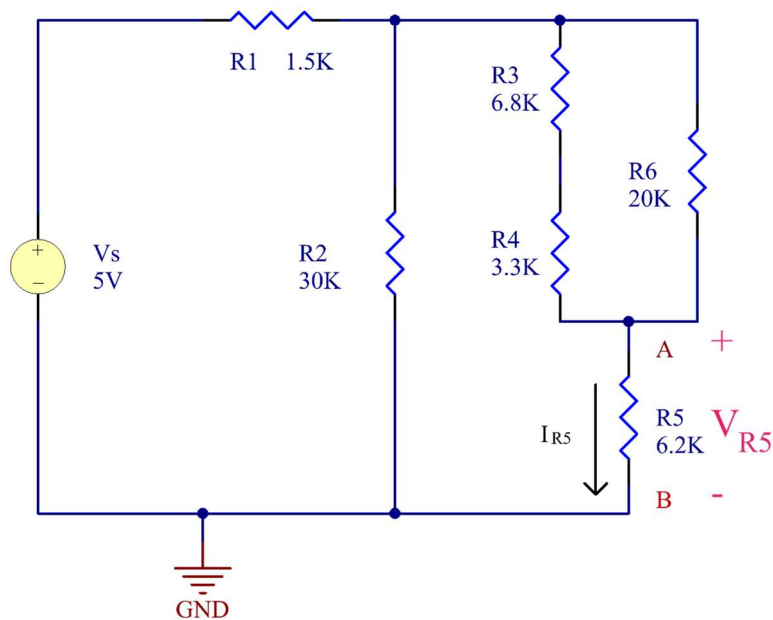


Figure 9

Using procedure explained in the theory section and/or your knowledge about finding a Thevenin equivalent, calculate the value of V_{TH} and R_{TH} for the circuit shown in Figure 9. Use R_5 as the load resistance (i.e. R_5 is removed for calculating the value of R_{TH} and V_{TH}). You can use the LTSpice simulator to check your Thevenin equivalents for this circuit if you want to. BTW LTSpice can not directly give you Thevenin equivalent values.

a. $V_{TH} = \underline{4.7619} \text{ V}$

$$V_{th} = (30000/31500) * 5 \text{ V}$$

b. $R_{TH} = \underline{8.1395} \text{ K}\Omega$

$$R_{346} = (1/10100 + 1/20000)^{-1} = 6710.963 \text{ ohms}$$

$$R_{12} = ((1/1500 + 1/30000)^{-1}) = 1428.571 \text{ ohms}$$

$$R_{th} = R_{346} + R_{12} = 8139.535 \text{ ohms}$$

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7. Calculate the value of I_{R5} using the Thevenin Equivalent Circuit

$$I_{R5} = V_{th} / (R_{th} + R_5) = 4.7619 / (8139.535 + 6200) = 0.332 \text{ mA}$$

a. $I_{R5} = \underline{0.332} \text{ mA}$

8. Implement the Thevenin Equivalent Circuit on the proto board and measure the voltage across $R5$ as shown in Figure 10. Pick the closest 5% resistor for R_{TH} .

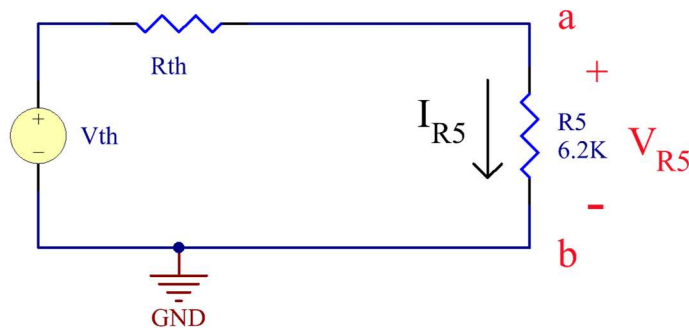


Figure 10. Thevenin equivalent of Circuit in Fig.8.

a. $V_{R5}(\text{Measured}) = \underline{2.0544} \text{ V}$, $I_{R5} = V_{R5} / R_5 = \underline{.3314} \text{ mA}$

- b. Does the value of I_{R5} match the calculated value in Part 7a?

It does, just about 1% error which is par for the course when comparing the same theoretical value using different calculation parameters.

Post Lab Questions

1. What is meant by the word "equivalent" in Thevenin Equivalent circuits?

In a Thevenin equivalent, "equivalent" means that from the load's point of view, the simplified circuit behaves exactly the same as the original circuit. Specifically, at the chosen terminals it produces the same voltage-current ($V-I$) relationship, so any load connected there will see the same terminal voltage and draw the same current. The original network can therefore be replaced by a single in series with R_{th} without changing the load's behavior.

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2. What is the practical value of Thevenin Equivalent circuits? Give several practical applications in which Thevenin Equivalent circuits are used.

The practicality lies in the simplistic nature of equivalent circuits. All you need to measure values that pertain to a component in a complex circuit is the equivalent resistance and voltage, it is much easier to construct a circuit consisting of one resistor and voltage source than an arbitrarily complex circuit.

Some practical examples:

- Max power transfer states that a load reaches max power at $R_L = R_{th}$, this is a direct application of thevenin Equivalents.
- Helpful to determine actual load driven by a battery, internal resistance droops voltage so modeling it as R_{th} is helpful.
- General troubleshooting and test simplification is something Thevenin lends itself to naturally.
- Outputs of amplifier stages are often modeled as a Thevenin source driving the next stage/load.